CS 237: PROBABILITY IN COMPUTING

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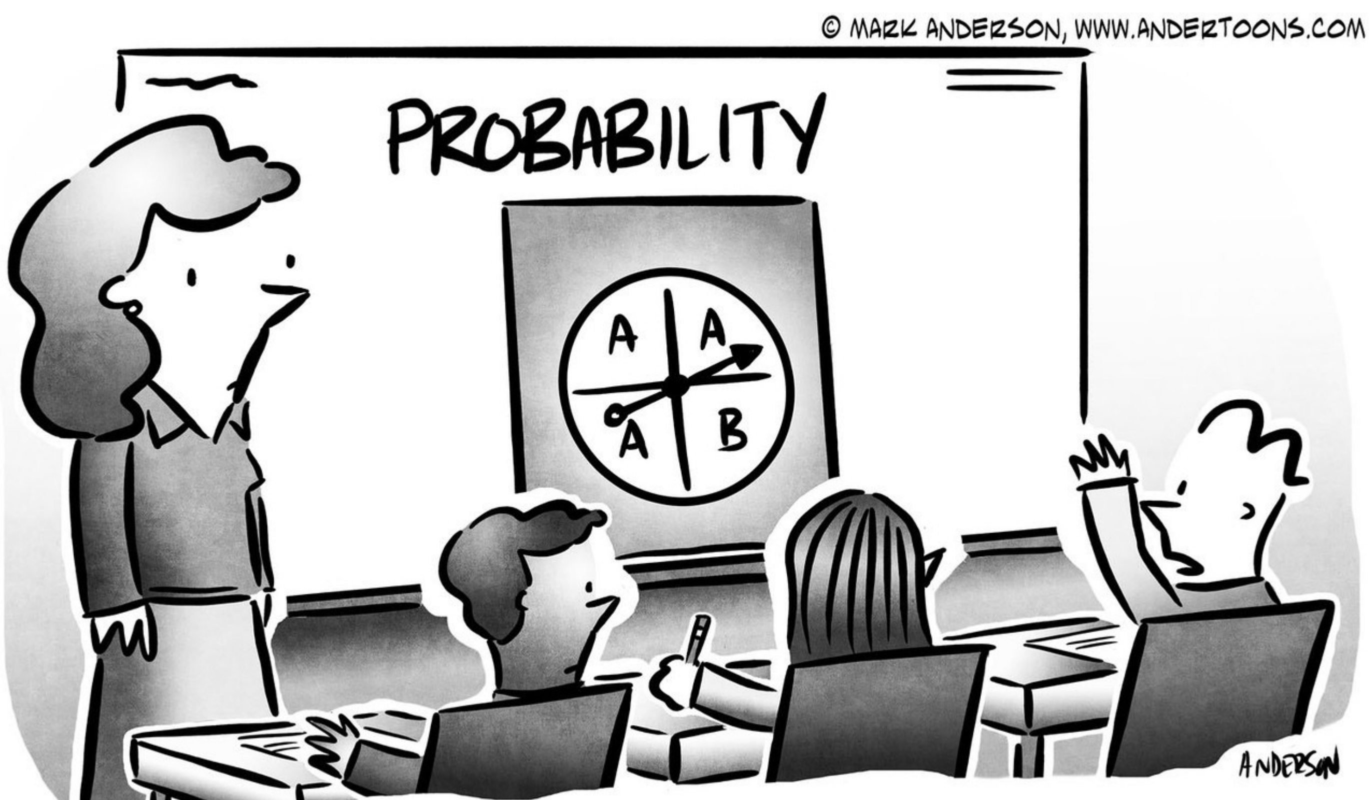

"I know mathematically that A is more likely, but I gotta say, I feel like B wants it more."

LECTURE 5

Last time

- More axioms and probability rules
- Continuous Probability Spaces
- Anomalies with Continuous Probability

Today

- Quiz 1
- More sample spaces
- Introduction to random variables

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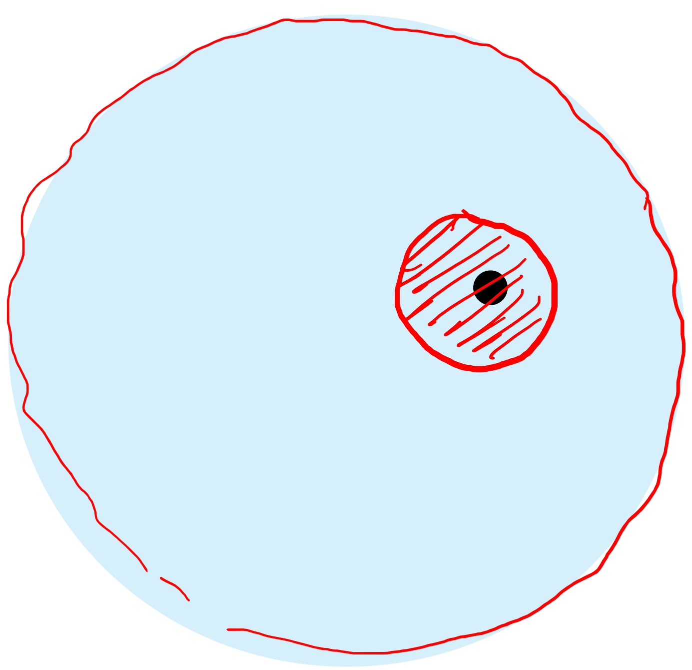
GET READY FOR QUIZ 1!

- Turn off your cellphone
- Put away all electronic devices
- Have a pen, pencil, and eraser ready
- Write your name and BU id at the top of the quiz
- When you finish, flip the quiz face down on your desk
- When the timer ends, hand the quiz to the instructor, a.k.a., me.

10 minutes to complete this Quiz

DARTS WITH A CIRCULAR BOARD

- Experiment: throw a dart at a circular board
- Outcome: hitting point p
- Sample space: the disk
- Probability function: $Pr(p \in P) = \dfrac{\text{area of } P}{\text{area of disk}}$



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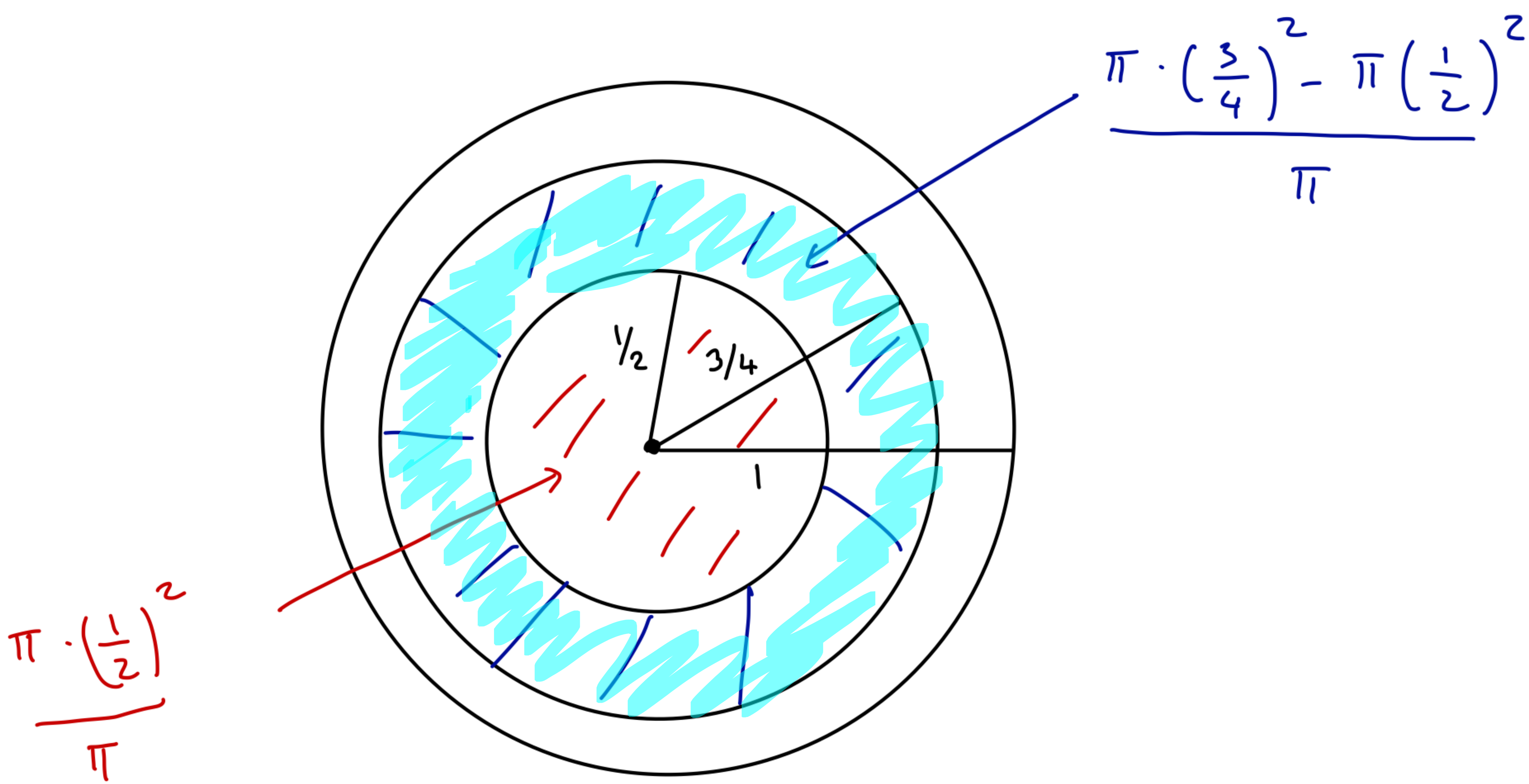
UNIFORM SAMPLING FROM THE UNIT DISK



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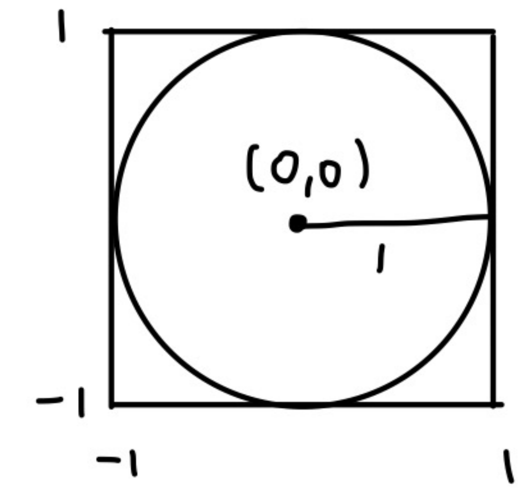
UNIFORM SAMPLING FROM THE UNIT DISK

Sample from the square enclosing the disk

If outside the disk, reject the point and repeat

Otherwise, return the sampled point




```
# uniform sampling from a unit disk centered at (0, 0)
def uniform_unit_disk():
    while True:
        # sample from the square
        (x, y) = uniform_rectangle(-1, 1, -1, 1)
        # check if (x, y) is on the disk
        if x**2 + y**2 <= 1:
            return (x, y)
```

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UNIFORM SAMPLING FROM THE UNIT DISK

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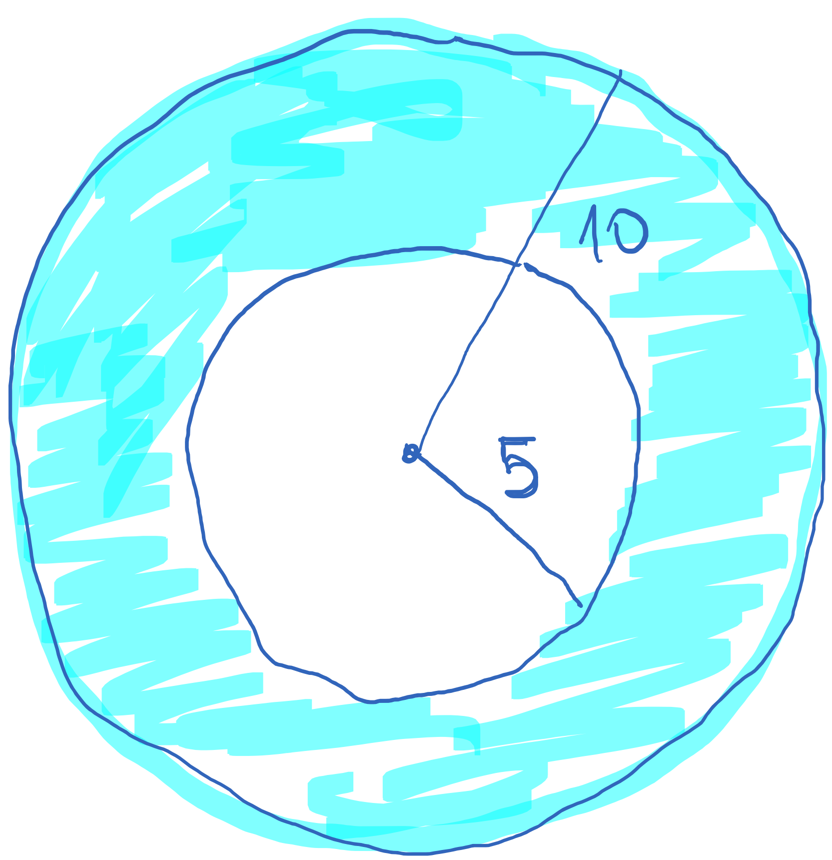
```

Think: how to sample from a disk of radius r?

DARTS WITH A CIRCULAR BOARD

(Math on the board)

- Charlie throws a dart at a circular board with radius 10
- What is the probability that the distance between the hitting point and the center of the board is **at least 5**?



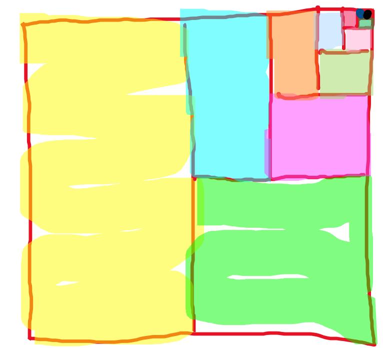
$$\Omega = \pi \cdot 10^2$$

$$E = \pi \cdot 10^2 - \pi \cdot 5^2$$

$$\Pr = \frac{\cancel{\pi} \cdot 10^2 - \cancel{\pi} \cdot 5^2}{\cancel{\pi} \cdot 10^2} = \frac{100-25}{100} = \frac{75}{100} = \frac{3}{4}$$

SUM OF GEOMETRIC SERIES: REVIEW

$$\sum_{k=1}^{\infty} \frac{1}{2^k} = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots = 1$$



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$$\sum_{k=0}^{\infty} ar^k = a + ar + ar^2 + \cdots = S$$

$$- \quad ar + ar^2 + ar^3 + \cdots = Sr$$

$$a = S - Sr \Rightarrow a = S(1-r) \Rightarrow S = \frac{a}{1-r}, \quad 0 \leq r < 1$$

COIN TOSSING REVISITED

- Experiment: toss a fair coin until we obtain heads
- Sample space: $\Omega = \{H, TH, TTH, \dots\} = \{T^i H : i \in \mathbb{N}\}$
- Probability function:

outcomes: $Pr(\{T^i H\}) = \frac{1}{2^{i+1}} \geq 0$

additivity

events: $Pr(S) = \sum_{\omega \in S} Pr(\{\omega\}) \geq 0$

- Question: does it satisfy the axioms?

$Pr(\Omega) = 1$

$Pr(\Omega) = \sum_{i=0}^{\infty} \frac{1}{2^{i+1}} = \sum_{i=1}^{\infty} \frac{1}{2^i} = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots = 1$

SAMPLE SPACES

- Discrete sample spaces: outcomes are listable
  - Toss a coin, roll two dice, toss a coin until we get heads, …
- Continuous sample spaces: outcomes are not listable
  - Spinner, darts

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SAMPLE SPACES

- Discrete sample spaces: outcomes are listable
  - We assign a probability to each outcome
  - The probability of an event is the sum of the probabilities of the outcomes comprising the event
- Continuous sample spaces: outcomes are not listable
  - Individual outcomes have probability zero
  - We assign probabilities to intervals and 2D regions

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SUMMARY: 5 TYPES OF SAMPLE SPACES

1. Finite Uniform Sample Spaces: Example $\Pr(A) = \frac{|A|}{|\Omega|}$

- Experiment: Roll a die, count the number of dots showing
- Sample space: $\Omega = \{ 1, \quad 2, \quad 3, \quad 4, \quad 5, \quad 6 \}$
- Probability Function:

$$P(1) = \frac{1}{6}, \qquad P(2) = \frac{1}{6}, \qquad P(3) = \frac{1}{6}$$

$$P(4) = \frac{1}{6}, \qquad P(5) = \frac{1}{6}, \qquad P(6) = \frac{1}{6}$$

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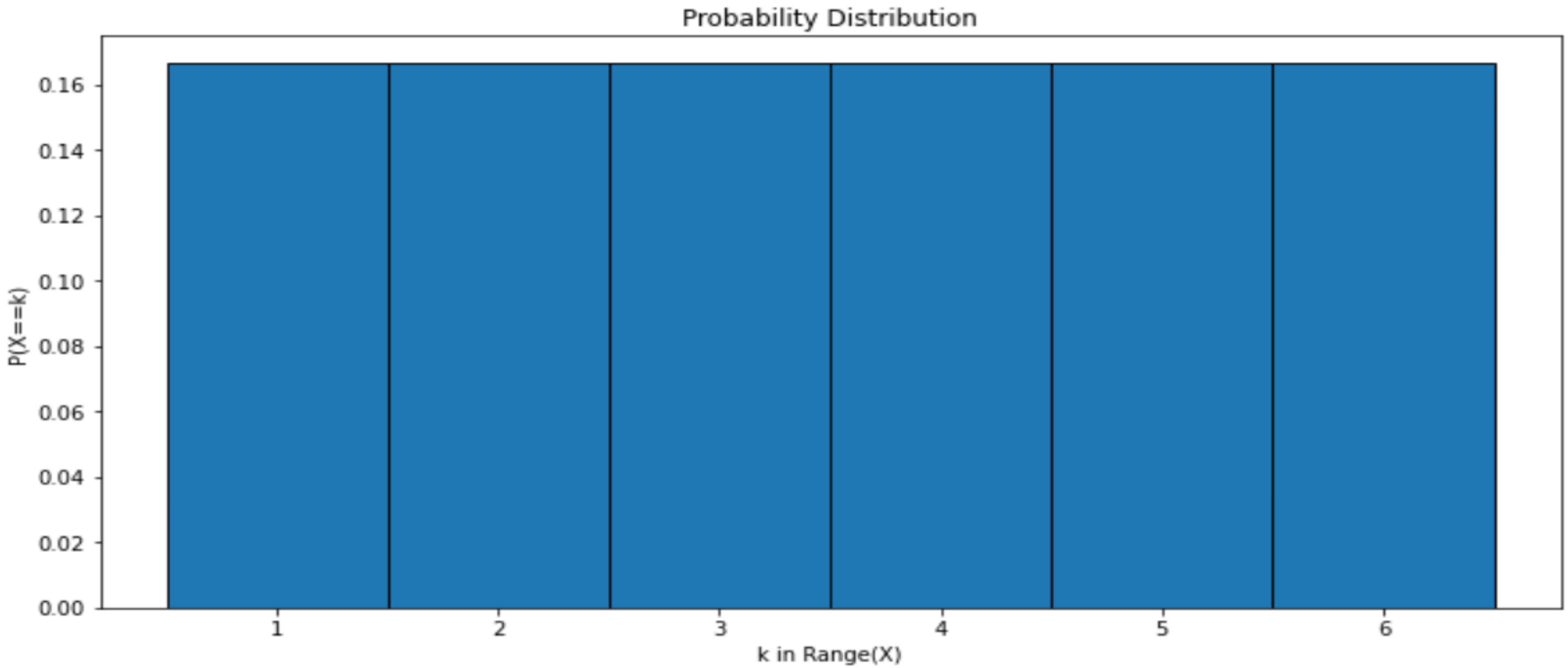
SUMMARY: 5 TYPES OF SAMPLE SPACES

1. Finite Uniform Sample Spaces: Example $\Pr(A) = \frac{|A|}{|\Omega|}$

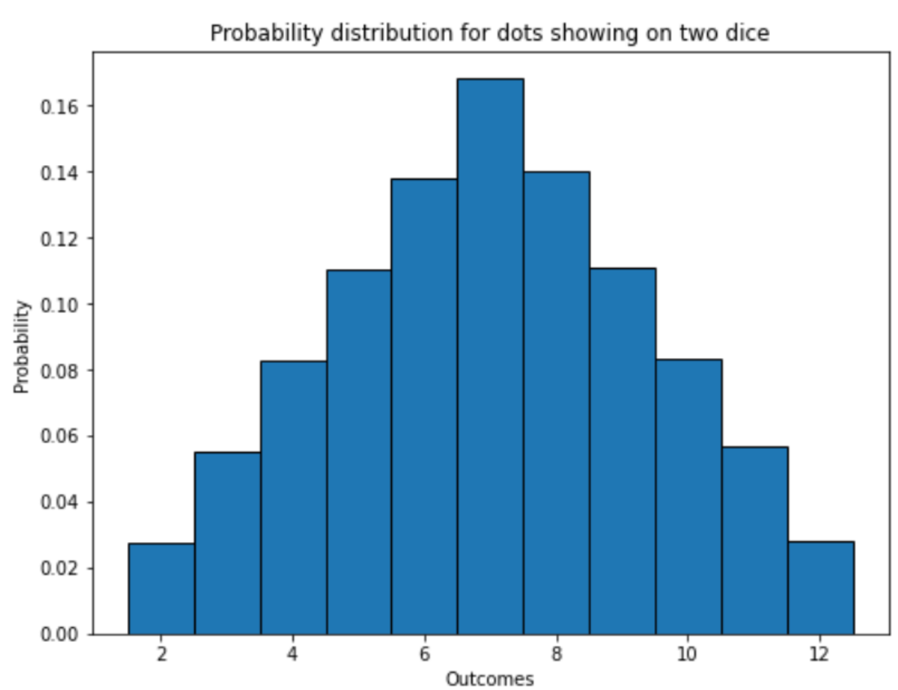


SUMMARY: 5 TYPES OF SAMPLE SPACES

2. Finite Non-Uniform Sample Spaces: Example

- Experiment: Throw two dice, count the number of dots showing
- Sample space: $\Omega = \{\, 2, 3, 4, \ldots, 12\}$
- Probability Function:

ω :	2	3	4	5	6	7	8	9	10	11	12
Pr(ω) :	1/36	2/36	3/36	4/36	5/36	6/36	5/36	4/36	3/36	2/36	1/36



	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

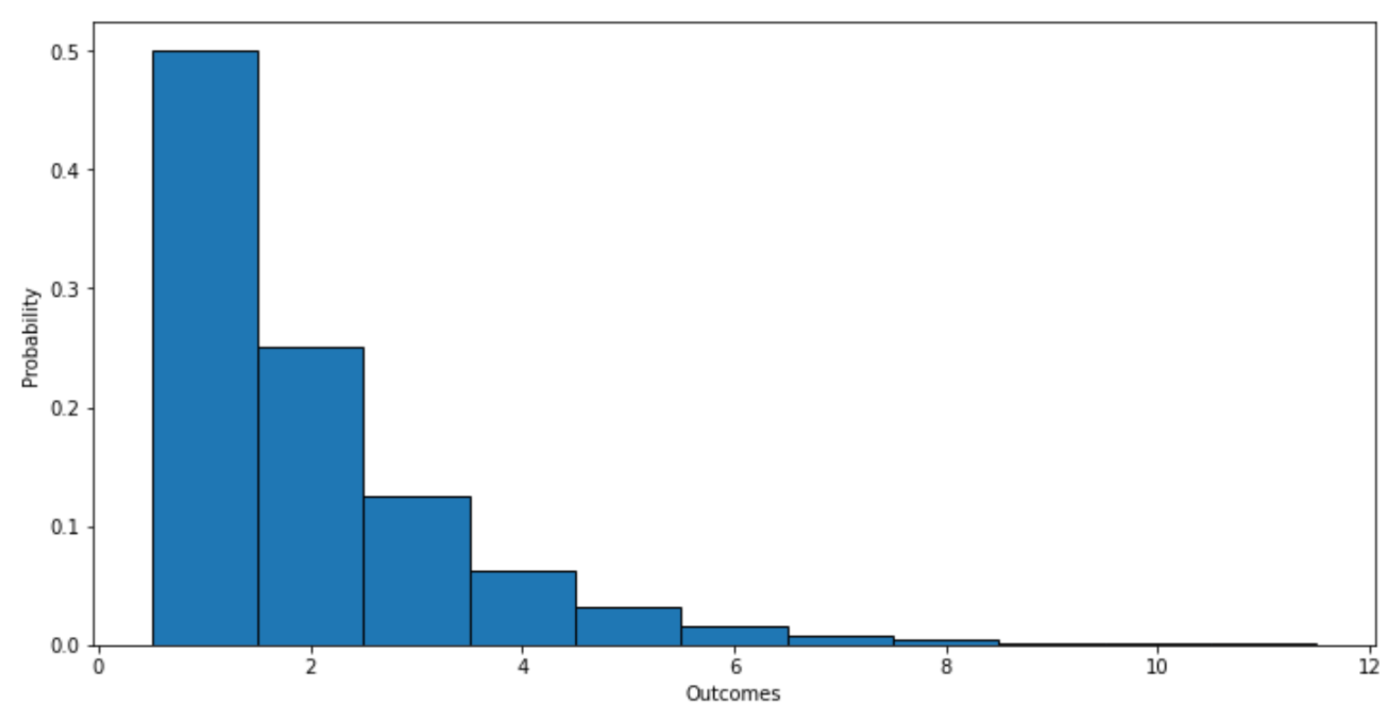
SUMMARY: 5 TYPES OF SAMPLE SPACES

$Pr(\{TH\}) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$

3. Countably Infinite Non-Uniform Sample Spaces: Example 1

- Experiment: Flip a coin until a head, count how many flips.
- Sample space: $\Omega = \{ 1, \quad 2, \quad 3, \quad 4, \quad 5, \quad .... \}$
- Probability Function:

ω :	1	2	3	4	...	k	...
Pr(ω) :	1/2	1/4	1/8	1/16	...	$1/2^k$	...



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SUMMARY: 5 TYPES OF SAMPLE SPACES

4. Uncountably Infinite Uniform Sample Spaces:

- Experiment: sample an interval uniformly from $[a, b]$
- Outcome: $[\mathrm{c}, \mathrm{d}] \subseteq [a, b]$
- Sample space: $\Omega = [a, b]$
- Probability function: for all $[c, d] \subseteq [a, b]$, we have

$$Pr(x \in [c, d]) = \frac{\text{length of } [c, d]}{\text{length of } [a, b]} = \frac{d - c}{b - a}$$

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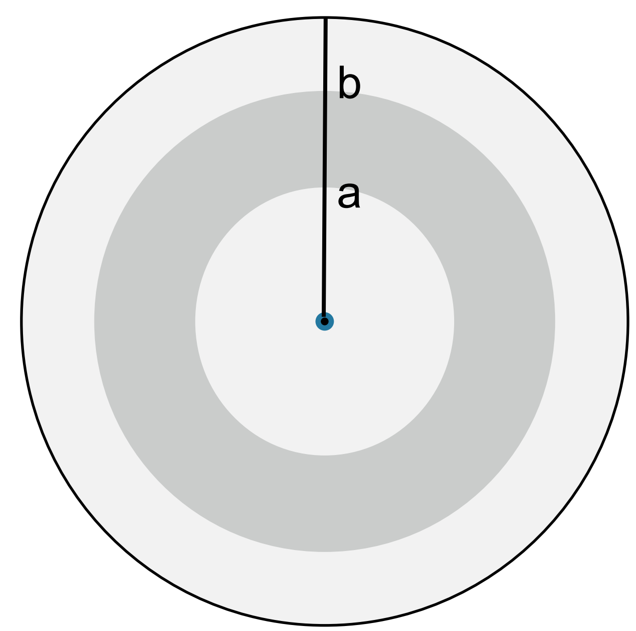
SUMMARY: 5 TYPES OF SAMPLE SPACES

5. Uncountably Infinite Non-Uniform Sample Spaces: Example

- Experiment: Throw a dart at a circular target of radius 1.0, what is the probability of hitting the annulus below?
- Sample space: $\Omega = [\ 0\ ..\ 1\ ]$
- Event: $\omega \in [a..b]$
- Probability Function:



$$\Pr(\,[a\,..\,b]\,) \;=\; \frac{\pi\, b^2 - \pi\, a^2}{\pi\, 1^2} \;=\; b^2 - a^2$$

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ASK THE AUDIENCE

Experiment: Pick a BU student uniformly at random and measure their height in meters, rounded to 2 decimal places.

The Sample Space is:

A. Finite and uniform

B. Finite and not uniform

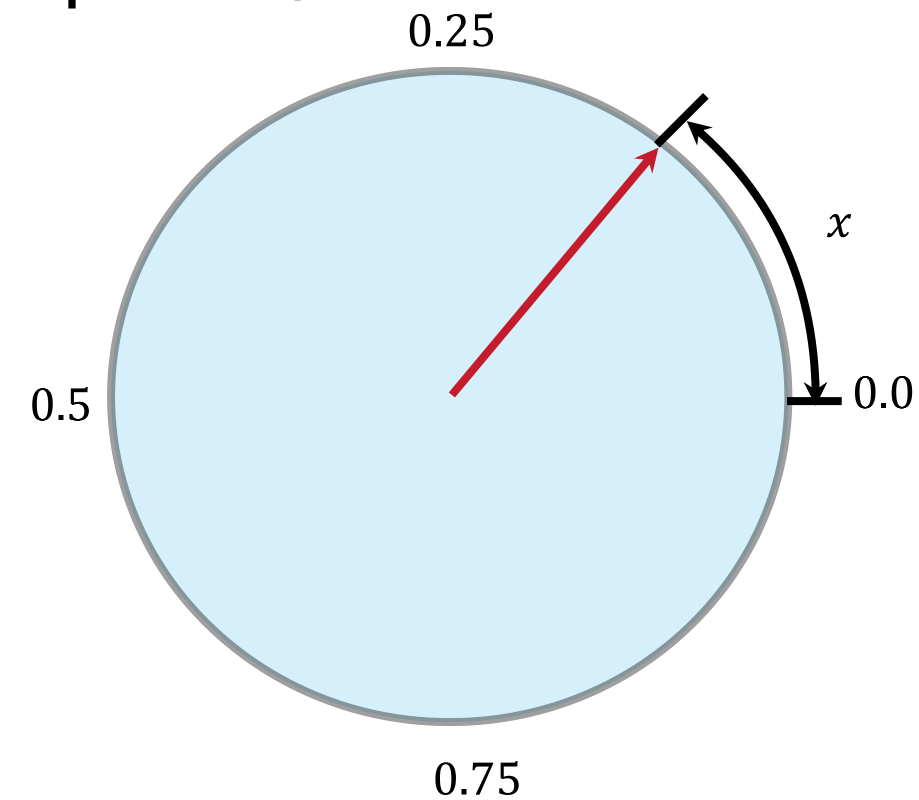
C. Countably infinite

D. Uncountably infinite and uniform

E. Uncountably infinite and not uniform

ASK THE AUDIENCE

What is the probability that when you spin the spinner, the arrow lands in the half-open interval [0,1/4) ?



A. ¼-epsilon

B. 1/4

C. The probability doesn't exist

$$Pr(x \in [0, 1/4)) = Pr(x \in [0, 1/4]) = 1/4 - 0 = 1/4$$

$$Pr(x \in [0, 1/4]) = Pr(x \in [0, 1/4) \cup \{4\})$$
$$= Pr(x \in [0, 1/4)) + Pr(x = \{4\})$$
$$= Pr(x \in [0, 1/4))$$

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ASK THE AUDIENCE

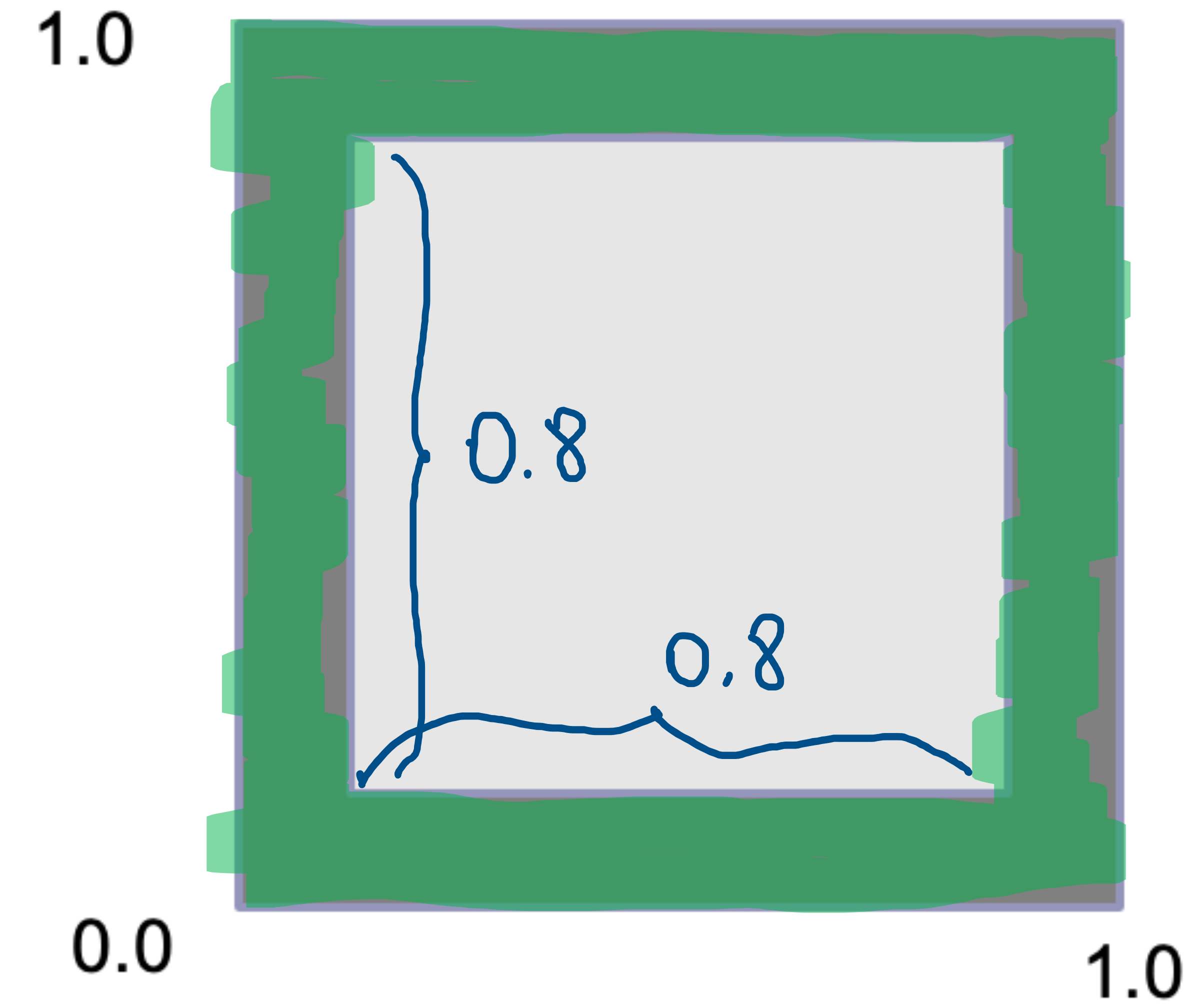
Suppose you throw a dart at a square target 1 meter on a side. What is the probability that it lands within 0.1 m of an edge?

A. $0.8^2 = 0.64$

B. $0.9^2 = 0.81$

C. $1 - 0.8^2 = 0.36$

D. $1 - 0.9^2 = 0.19$



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21

ASK THE AUDIENCE: PROBABILITY RULES

- We roll a die four times
- What is the probability that we obtain exactly one 6?

A. $1 - \left(\frac{5}{6}\right)^4$

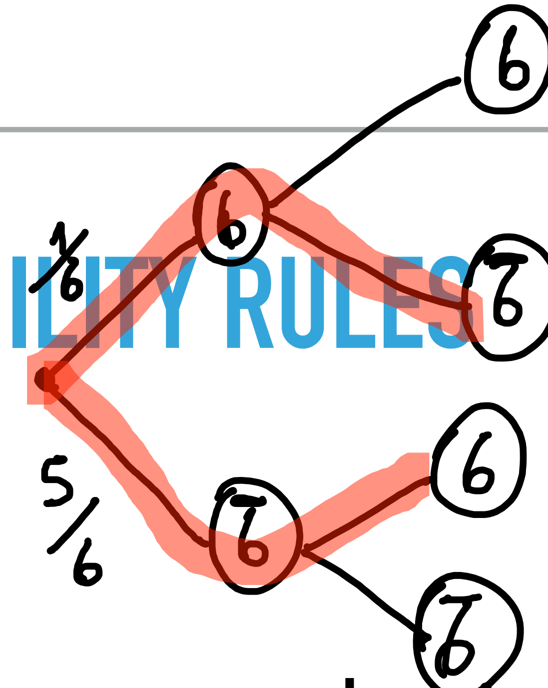
B. $\frac{4}{6}$

C. $\frac{1}{6}$

D. $4 \cdot \frac{1}{6} \cdot \left(\frac{5}{6}\right)^4$

E. None of the above

ASK THE AUDIENCE: PROBABILITY RULES



- We roll a die four times
- What is the probability that we obtain exactly one 6?

$6\ \bar{6}\ \bar{6}\ \bar{6}$

$\bar{6}\ 6\ \bar{6}\ \bar{6}$

$\bar{6}\ \bar{6}\ \bar{6}\ 6$

$\bar{6}\ \bar{6}\ \bar{6}\ 6$

Pr[roll a die once and get one six] = 1/6

Pr[roll a die twice and get one six] $= \frac{1}{6}\cdot\frac{5}{6} + \frac{5}{6}\cdot\frac{1}{6}$

$= 2\cdot\frac{1}{6}\cdot\frac{5}{6}$

⋮

Pr[roll a die four times and get one six] $= 4\cdot\frac{1}{6}\cdot\left(\frac{5}{6}\right)^3$

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PROBABILITY RULES

- Alice, Barbara, and Charlotte are the three candidates running for president of BU Girls Who Code. One of these three will always win.
- Suppose that Alice and Barbara have the same probability of winning, and the probability that Charlotte wins is only 1/2 of that of either Alice or Barbara
- Find the probability that Alice or Charlotte wins

$A = \Pr[\text{Alice wins}]$, $B = \Pr[\text{Barbara wins}]$, $C = \Pr[\text{Charlotte wins}]$

$A = B$, $C = A/2$, $1 = A + B + C \Rightarrow 1 = 2A + A/2 \Rightarrow A = \frac{2}{5} = B$

$\Pr[\text{Alice or Charlotte wins}] = A + C = \frac{2}{5} + \frac{1}{5} = 3/5$

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ASK THE AUDIENCE: PROBABILITY RULES AND SYMMETRY

- We shuffle a standard deck of cards. Each permutation is equally likely
- Suppose we deal 13 cards
- What is the probability that the last card we dealt is an ace?

(A) $\frac{1}{52}$ (B) $\frac{1}{13}$ (C) $\frac{\binom{51}{12}\cdot\binom{4}{1}}{\binom{52}{13}\cdot 13}$ (D) Both B and C are correct

$$\frac{\binom{4}{1}\binom{51}{12}12!}{\binom{52}{13}13!} = \frac{4\cdot\frac{\cancel{51\cdot 50\cdots 40}}{\cancel{12!}}}{\frac{52\cdot\cancel{51\cdots 40}\cdot\cancel{13}}{\cancel{13}\cdot\cancel{12!}}} = \frac{4}{52} = \frac{1}{13}$$

ASK THE AUDIENCE

- We shuffle a standard deck of cards. Each permutation is equally likely
- Suppose we deal 13 cards
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(A) $\frac{1}{52}$ (B) $\frac{1}{13}$ (C) $\frac{\binom{51}{12}\cdot\binom{4}{1}}{\binom{52}{13}\cdot 13}$ (D) Both B and C are correct

By symmetry:

$\Pr(13^{th}\text{ card is Ace}) = \Pr(13^{th}\text{ card is King}) = \Pr(13^{th}\text{ card is }4)$

$\vdots$

$\frac{4}{13}$

A | 1 | 2 | 3 | 4 | 5 | 6 | 7 | ... | J | Q | K

ASK THE AUDIENCE: PROBABILITY RULES AND SYMMETRY

- We shuffle a standard deck of cards. Each permutation is equally likely
- Suppose we deal 2 cards
- What is the probability that the second card is higher in rank than the first card?

Consider these three options

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{8}{17}$ (D) $\frac{13}{25}$

- 1st > 2nd
- 2nd > 1st
- 1st = 2nd → $\frac{\binom{13}{1}\cdot\binom{4}{2}}{\binom{52}{2}} = \frac{\frac{\cancel{13\cdot 4}\cdot 3}{\cancel{2}}}{\frac{\cancel{52}\cdot 51}{\cancel{2}}} = \frac{3}{51}$

same → $\frac{1}{2} - \frac{3}{51} = \frac{8}{17}$

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58

ASK THE AUDIENCE

We toss a coin 5 times. The outcome is a string of length 5 of H's and T's, where H means ``heads'' and T means ``tails''.

How many outcomes are there where the first toss is H **or** we have exactly 2 T's?

A. 6
B. 10
C. 16
D. 20
E. None of the above